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1. Age Calculator

Algorithm:

- Step 1: Start
- Step 2: Set current year = 2026
- Step 3: Input birthyear from the user
- Step 4: Calculate age = current year - birth year
- Step 5: Print age
- Step 6: Stop

2. Area Calculator

Algorithm:

- Step 1: Start
- Step 2: Set pi = 3.14
- Step 3: Input radius of the circle
- Step 4: Calculate area circle = $\pi * \text{radius}^2$
- Step 5: Print area circle
- Step 6: Input length and width of the rectangle
- Step 7: Calculate area rectangle = length * width
- Step 8: Print area rectangle
- Step 9: Calculate total area = area circle + area rectangle
- Step 10: Print total area
- Step 11: Stop

3. Average Speed Calculator

Algorithm:

- Step 1: Start
- Step 2: Input distance1 and time1 for section 1

Step 3: Input distance2 and time2 for section 2
Step 4: Calculate $\text{speed1} = \text{distance1} / \text{time1}$
Step 5: Calculate $\text{speed2} = \text{distance2} / \text{time2}$
Step 6: Calculate total distance = distance1 + distance2
Step 7: Calculate total time = time1 + time2
Step 8: Calculate average speed = total distance / total time
Step 9: Print speed1, speed2, total distance, total time, average speed
Step 10: Stop

4. Bank Deposit Calculator

Algorithm:

Step 1: Start
Step 2: Input principle amount
Step 3: Calculate interest = principle * 0.07
Step 4: Print interest
Step 5: Calculate total = principle + interest
Step 6: Print total amount after one year
Step 7: Stop

5. Compound Expression

Algorithm:

Step 1: Start
Step 2: Input three numbers: a, b, c
Step 3: Calculate sum = $a + b + c$
Step 4: Print sum
Step 5: Calculate average = $\text{sum} / 3$
Step 6: Print average

Step 7: Calculate expression = $(a + b)^2 + (b + c)^2$

Step 8: Print expression

Step 9: Stop

6. Data Storage Conversion

Algorithm:

Step 1: Start

Step 2: Input data size in MB

Step 3: Calculate $kb = size * 1024$

Step 4: Print size in KB

Step 5: Calculate $GB = size / 1024$

Step 6: Print size in GB

Step 7: Stop

7. Electricity Bill Calculator

Algorithm:

Step 1: Start

Step 2: Input unit consumed

Step 3: Set charge per unit = 12

Step 4: Set service charge = 100

Step 5: Calculate bill amount = unit consumed * charge per unit

Step 6: Calculate total bill = bill amount + service charge

Step 7: Print total bill

Step 8: Stop

8. Exam Result Sheet

Algorithm:

Step 1: Start
Step 2: Input name and marks for subject 1
Step 3: Input name and marks for subject 2
Step 4: Input name and marks for subject 3
Step 5: Input name and marks for subject 4
Step 6: Calculate total marks = mark1 + mark2 + mark3 + mark4
Step 7: Print total marks
Step 8: Calculate avg marks = total marks / 4
Step 9: Print avg marks
Step 10: Calculate percentage = (total marks / 400) * 100
Step 11: Print percentage
Step 12: Stop

9. Mathematical Expression

Algorithm:

Step 1: Start
Step 2: Input four numbers: a, b, c, d
Step 3: Calculate expression = $(a + b)^2 + (b + c)^2 + (c + d)^2$
Step 4: Print expression
Step 5: Calculate average = $(a + b + c + d) / 4$
Step 6: Print average
Step 7: Stop

10. Monthly Expense Calculator

Algorithm:

Step 1: Start
Step 2: Input expenses for food, clothes, education, transport, others

Step 3: Calculate monthly expenses = food + clothes + education + transport + others

Step 4: Print monthly expenses

Step 5: Calculate yearly expenses = monthly expenses * 12

Step 6: Print yearly expenses

Step 7: Stop

11. Physics Formula

Algorithm:

Step 1: Start

Step 2: Input initial velocity (u), final velocity (v), and time (t)

Step 3: Calculate acceleration = $(v - u) / t$

Step 4: Print acceleration

Step 5: Calculate distance = $u * t + 0.5 * \text{acceleration} * t^2$

Step 6: Print distance

Step 7: Stop

12. Profit Margin Calculator

Algorithm:

Step 1: Start

Step 2: Input product1 name, cost1, sell1

Step 3: Calculate profit1 = sell1 - cost1 and print profit1

Step 4: Input product2 name, cost2, sell2

Step 5: Calculate profit2 = sell2 - cost2 and print profit2

Step 6: Input product3 name, cost3, sell3

Step 7: Calculate profit3 = sell3 - cost3 and print profit3

Step 8: Calculate total cost = cost1 + cost2 + cost3

Step 9: Calculate total selling = sell1 + sell2 + sell3

Step 10: Calculate total profit = total selling - total cost

Step 11: Calculate profit percentage = (total profit / total cost) * 100

Step 12: Print total cost, total selling, total profit, profit percentage

Step 13: Stop

13. Salary Calculator

Algorithm:

Step 1: Start

Step 2: Input basic salary

Step 3: Calculate HRA = basic * 0.10

Step 4: Print HRA

Step 5: Calculate DA = basic * 0.05

Step 6: Print DA

Step 7: Calculate gross salary = basic + HRA + DA

Step 8: Print gross salary

Step 9: Stop

14. Speed Calculator

Algorithm:

Step 1: Start

Step 2: Input distance in km

Step 3: Input time in hours

Step 4: Calculate speed = distance / time

Step 5: Print speed in km/h

Step 6: Stop

15. Travel Cost Calculator

Algorithm:

Step 1: Start

Step 2: Input distance to travel in km

Step 3: Input expenses per km

Step 4: Input number of days

Step 5: Calculate total cost = distance * expenses * days

Step 6: Print total cost

Step 7: Stop